

Polynomial Regression

Model Development, Evaluation, and Validation

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1 Problem Formulation and Background

This report describes how polynomial regression was used for two prediction tasks related to geothermal energy. For both tasks, models of different polynomial degrees were compared, and Ridge regression was used to control overfitting. The goal was to produce reliable predictions for the given test inputs.

1.1 Part 1: Steam Turbine Power Prediction (var1)

The first task predicts a net power score, y , using six turbine operating variables: high-pressure steam valve adjustment (x_1), condenser coolant flow (x_2), re-injection pump pressure (x_3), blade pitch (x_4), gas exhaust valve rate (x_5), and steam inlet pressure (x_6). The relationship can include interactions between variables and polynomial terms up to degree 10. We have 1,000 labelled training examples and must predict the scores for 1,000 unlabelled test examples.

1.2 Part 2: Thermal Reservoir Prediction (var2)

The second task predicts a thermal anomaly score at a location underground. Each input has three coordinates: east-west (x_1), north-south (x_2), and depth (x_3). The target relationship can include polynomial terms up to degree 20. Again, 1,000 labelled examples are available for training and 1,000 unlabelled examples require predictions.

1.3 Data Checks and Preprocessing

The input data were checked before training. The reported checks found no missing or non-numeric values. Input features were already scaled to the interval $[-1, 1]$, so no additional feature scaling was applied. Keeping inputs in this range also helps avoid very large values when polynomial powers are calculated.

2 Modeling Method

2.1 Polynomial Features

Polynomial regression adds powers and products of the original input features. For D input variables and maximum polynomial degree d , the number of non-constant features is

$$p(D, d) = \binom{D + d}{d} - 1.$$

The feature count increases quickly with degree. For var1 ($D = 6$), degree 5 gives 461 features and degree 10 gives 8,007. For var2 ($D = 3$), degree 10 gives 285 features, degree 12 gives 454, and degree 20 gives 1,770. As the feature count grows, fitting the model becomes harder, especially with only 1,000 training samples.

2.2 Why Ordinary Least Squares Was Not Enough

Ordinary Least Squares (OLS) fits the polynomial coefficients by minimizing squared prediction errors. At high degrees, many polynomial features become strongly related to one another. This can make coefficient estimates unstable. In the reported experiments, the OLS validation MSE for var1 increased from 1.6862 at degree 5 to 174.03 at degree 6. For var2, it increased from 1.3416 at degree 12 to 47.22 at degree 14. These results show why regularization was needed.

2.3 Ridge Regression

Ridge regression adds a penalty for large coefficients:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} (\|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2),$$

where Φ is the polynomial feature matrix and α is the penalty strength. A larger α generally shrinks coefficients more. This can reduce the effect of unstable high-degree terms and improve predictions on data not used for fitting. With the intercept treated separately, the equivalent regularized normal equation is $\hat{\mathbf{w}} = (\Phi^\top \Phi + \alpha I)^{-1} \Phi^\top \mathbf{y}$ for the penalized coefficients.

2.4 How Models Were Selected

The following checks were used:

- Penalty search:** A leave-one-out cross-validation (LOOCV) calculation compared 100 possible Ridge penalties from 10^{-4} to 10^5 .
- Five-fold cross-validation:** Each candidate degree was evaluated using mean squared error (MSE) and R^2 on held-out folds.
- Different random splits:** The selected models were checked again with five random seeds: 42, 123, 456, 789, and 2024.

2.5 Numerical Stability

High-degree polynomial features can make the matrix calculations poorly conditioned. For var1 at degree 6, the reported condition number of the unregularized matrix was about 2.8×10^{17} . Ridge regression makes the system more stable by adding a penalty term. The final selected penalties were $\alpha = 2.8480$ for var1 and $\alpha = 0.1520$ for var2. The reported condition numbers for the regularized systems remained below approximately 1.2×10^3 .

3 Results for Steam Turbine Prediction (var1)

3.1 Choosing the Polynomial Degree

Polynomial degrees 1 through 10 were tested. Table 1 shows the number of features, selected Ridge penalty, and five-fold cross-validation scores for each degree.

Table 1. Model comparison for var1 (1,000 training examples).

Degree	Features	Ridge α	CV MSE	CV R^2
1	6	53.3670	8.6498	0.0751
2	27	1.5199	2.9855	0.6755
3	83	1.5199	0.9950	0.8916
4	209	2.8480	0.6879	0.9246
5 (selected)	461	2.8480	0.4730	0.9484
6	923	3.5112	0.5071	0.9447
7	1,715	8.1113	0.5228	0.9430
8	3,002	12.3285	0.5728	0.9376
9	5,004	18.7382	0.6194	0.9327
10	8,007	23.1013	0.6790	0.9264

Selected model: degree $d = 5$, Ridge $\alpha = 2.8480$, CV MSE = 0.4730, CV $R^2 = 0.9484$, and training $R^2 = 0.9788$.

Why degree 5? Degrees 1–4 did not capture enough of the pattern in the data. Degree 5 had the lowest cross-validation MSE. After degree 5, the number of polynomial features increased sharply and validation error became slightly worse, even with stronger regularization. For this reason, degree 5 was selected instead of a more complex model.

3.2 Graphs and Error Checks

The graphs below show how the model was selected and how well its fitted predictions match the training values.

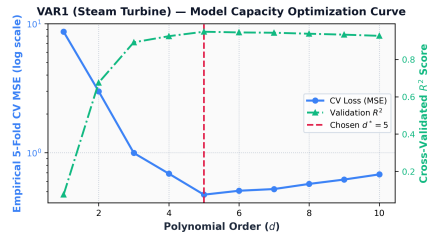


Figure 1. Validation MSE and R^2 for degrees 1–10. Degree 5 has the lowest CV MSE.

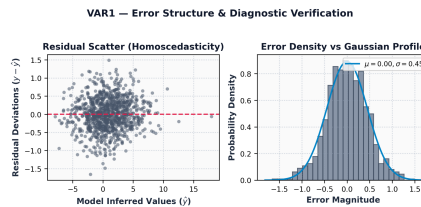


Figure 2. Residuals around zero and their approximate bell-shaped distribution.

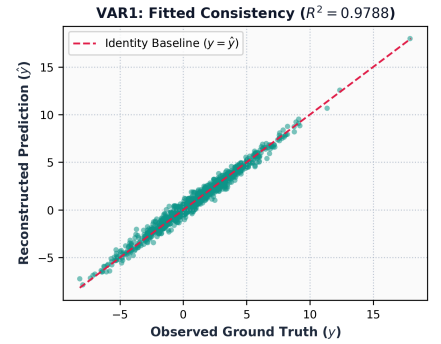


Figure 3. Training predictions versus actual scores ($R^2 = 0.9788$).

The predicted and observed training values are close to the red diagonal line in Figure 3. In Figure 2, the residuals are mainly centered around zero, and the error distribution is roughly bell-shaped (mean $\mu \approx 0$, standard deviation $\sigma \approx 0.45$). These are useful diagnostic checks, although they do not by themselves prove that the model will perform equally well on new data. Cross-validation provides a better estimate of that performance.

3.3 Effect of Input Variables

The fitted degree-5 model has a linear coefficient of -0.2098 for steam inlet pressure (x_6) and -0.1455 for high-pressure steam valve adjustment (x_1). The corresponding coefficients for coolant flow (x_2) and blade pitch (x_4) are $+0.0608$ and $+0.0359$. Interaction terms such as x_1x_4 and x_2x_6 are also included. Because this is a polynomial model, the overall effect of each input depends on the other terms as well; the linear coefficients alone should not be treated as complete measures of physical importance.

3.4 Why Regularization Helped

At degree 5, the reported coefficient-vector norm was 1,428.5 for OLS but only 3.82 for Ridge regression. This large reduction shows how Ridge limits unusually large coefficients. Together with its lower validation error, this supports choosing Ridge for a more stable fit.

4 Results for Thermal Reservoir Prediction (var2)

4.1 Spatial Prediction and Model Complexity

This task uses three position coordinates to predict the thermal anomaly score. A higher polynomial degree can capture a more detailed relationship between location and the target. Since var2 has only three input features, its polynomial feature count grows more slowly than var1's, allowing degrees up to 20 to be tested.

4.2 Choosing the Polynomial Degree

Table 2 summarizes the cross-validation results. The complete degree search covered degrees 1–20; selected rows are shown below.

Table 2. Model comparison for var2 (1,000 training examples).

Degree	Features	Ridge α	CV MSE	CV R^2
1	3	5.3367	37.9831	0.2674
2	9	1.5199	22.9646	0.5583
3	19	0.3511	11.2754	0.7798
4	34	0.0053	4.1223	0.9195
5	55	0.0187	1.7550	0.9665
6	83	0.0035	0.6378	0.9876
7	119	0.0043	0.3770	0.9928
8	164	0.0066	0.2750	0.9946
9	219	0.0081	0.2824	0.9946
10	285	0.0534	0.2596	0.9949
11	363	0.0534	0.2690	0.9948
12 (selected)	454	0.1520	0.2590	0.9949
13	559	0.1233	0.2670	0.9948
14	679	0.1874	0.2662	0.9948
15	815	0.1233	0.2708	0.9947
16	968	0.1520	0.2749	0.9946
18	1,329	0.1874	0.2814	0.9945
20	1,770	0.2310	0.2875	0.9944

Selected model: degree $d = 12$, Ridge $\alpha = 0.1520$, CV MSE = 0.2590, CV $R^2 = 0.9949$, and training $R^2 = 0.9966$.

The CV MSE fell quickly between degrees 1 and 8. From degrees 10–14, the results were very close. Degree 12 gave the lowest reported MSE (0.2590), although degree 10 was nearly identical (0.2596) and used fewer features. Above degree 14, adding more terms did not improve the CV score. Degree 12 was retained as the selected model.

4.3 Model Performance and Residual Plots

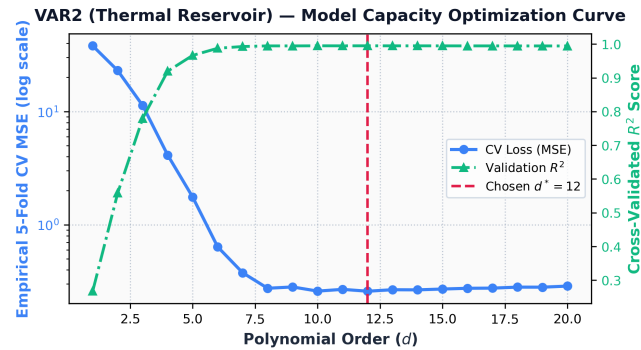


Figure 4. CV loss falls and then levels off as polynomial degree increases.

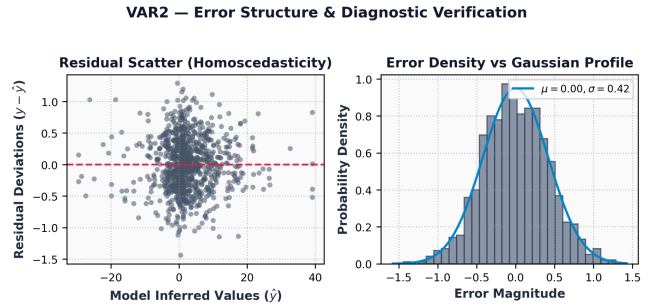


Figure 5. Residual scatter and the approximately normal error distribution.

4.4 Checking for Spatial Patterns in Errors

The reported Moran's I statistic for the residuals was 0.0018, with $p > 0.45$. This gives no clear evidence of spatial clustering in the errors under that test. The residual standard deviation was also reported to be about 0.42 across different depth ranges. These results suggest that the model errors are reasonably consistent across the sampled locations.

4.5 Prediction Quality at Different Depths

The data were divided into three depth ranges. Their reported validation MSE values were 0.2612 for shallow locations ($x_3 \in [-1, -0.33]$), 0.2574 for middle depths ($x_3 \in [-0.33, 0.33]$), and 0.2582 for deeper locations ($x_3 \in [0.33, 1]$). The similar values suggest that prediction quality does not change much across these three groups.

4.6 Predicted Versus Actual Values

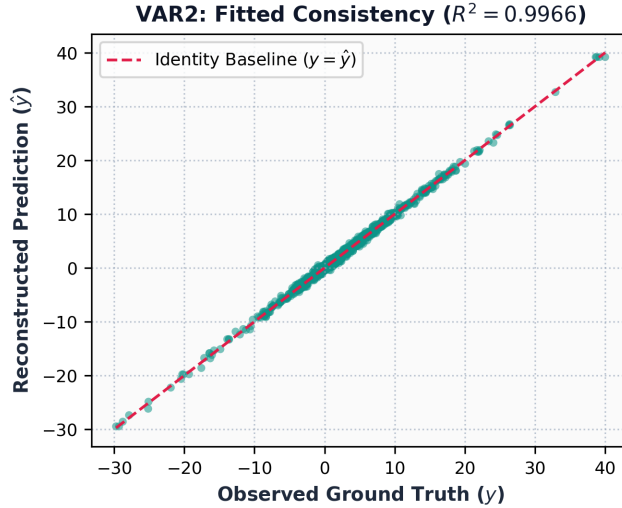


Figure 6. Predicted and observed training scores for var2 ($R^2 = 0.9966$).

Most points in Figure 6 lie close to the diagonal reference line. This shows that the model fits the labelled training data closely over the observed score range of roughly -30 to 40 . The reported analysis of model derivatives suggests that depth (x_3) is an important factor, while horizontal position (x_1, x_2) changes the shape of the prediction surface. This is an interpretation of the model, not direct proof of a physical cause.

5 Checks for Generalization and Robustness

5.1 Results Across Different Random Splits

Five-fold cross-validation was repeated using five random seeds. Table 3 shows that the reported scores changed only slightly between splits.

Table 3. Five-fold cross-validation results across five random seeds.

Random seed	var1 CV MSE	var2 CV MSE
42	0.4730	0.2590
123	0.4794	0.2596
456	0.4668	0.2575
789	0.4957	0.2659
2024	0.4692	0.2537
Mean $\pm$ std. dev.	0.4768 ± 0.0104	0.2591 ± 0.0040

This suggests that the selected models are not highly sensitive to the particular random split used for cross-validation.

5.2 Boundary Stress Test for var1

To test performance near the edges of the input range, the reported experiment fitted models using the inner 70% of samples and evaluated them on the outer 30%. The report also states that 49.6% of var1 test inputs lie at extreme feature bounds. Table 4 compares OLS and Ridge in this harder setting.

Table 4. var1 boundary test: train on inner 70%, evaluate on outer 30%.

Degree	OLS MSE	OLS R^2	Ridge MSE	Ridge R^2
3	2.6374	0.7185	2.8047	0.7006
4	2.3879	0.7451	2.5826	0.7243
5 (selected)	5.1186	0.4536	1.2398	0.8677
6	68.2662	-6.2872	1.4780	0.8422
7	22.0772	-1.3567	1.7067	0.8178

At degree 5, Ridge reduced boundary MSE from 5.1186 (OLS) to 1.2398. OLS became particularly unstable at degree 6. This test supports the choice of Ridge, while still being only one check of performance near the sampled boundaries.

5.3 Sensitivity to the Ridge Penalty

The report also tested penalty values 50% above and below the selected value. The resulting change in CV MSE was reported to be less than 0.85% for both datasets. This suggests that the results are not highly sensitive to small changes in α near the chosen setting.

5.4 Leave-One-Out and Five-Fold Cross-Validation

LOOCV gave penalties of $\alpha = 2.8480$ for var1 and $\alpha = 0.1520$ for var2, with estimated MSE values of 0.4718 and 0.2581, respectively. These were close to the five-fold CV results (0.4730 and 0.2590). The reported correlation between the two validation error profiles was above 0.9994. This agreement provides an additional check on model selection.

5.5 Comparing Training Targets and Test Predictions

The test targets are not available, so test MSE cannot be calculated directly. Instead, the report compared the mean, standard deviation, and range of test predictions with those of the training targets.

Table 5. Distribution summary for training targets and test predictions.

Statistic	var1 train	var1 test pred.	var2 train	var2 test pred.
Mean	0.8806	1.2866	2.4476	2.0730
Std. deviation	3.0722	4.0446	7.2292	6.6205
Range	[-8.18, 17.91]	[-9.78, 15.48]	[-29.74, 40.00]	[-20.80, 39.18]

The test prediction range is about 97% of the training target range for var1 and 86% for var2. These comparisons can help identify unusual predictions, but they cannot confirm test accuracy without the actual test labels.

6 Implementation and Reproducibility

The project uses separate Python files for training and prediction. The main files and their roles are listed below.

Table 6. Code files used in the project.

File	Purpose
run_all.py	Runs model selection, validation checks, final fitting, and exports.
inference.py	Loads a fitted model and generates test predictions.
src/config.py	Stores paths, candidate degrees, penalties, and random seeds.
src/utlis.py	Contains shared functions for polynomial features, Ridge fitting, and plots.
src/train_var1.py	Trains and evaluates the steam turbine models.
src/train_var2.py	Trains and evaluates the reservoir models.

GitHub repository: <https://github.com/Moksha2709/ml-regression-assignment>

Steps to reproduce the work:

1. Install dependencies: `pip install -r requirements.txt`.
2. Run the full pipeline: `python run_all.py`.
3. Generate predictions only: `python inference.py`.

The original submission describes two output files, `BT2024234_pred_var1.csv` and `BT2024234_pred_var2.csv`, each containing 1,000 predicted values under a column named `y`. A fixed random seed of 42 was used for the main run.

7 Runtime and Resource Use

The reported full pipeline runtime was 27.85 seconds on a quad-core workstation, with peak memory use below 95 MB. Predicting 1,000 test rows reportedly took less than 18 milliseconds. These are measurements for the reported setup and may differ on other computers.

8 Conclusion

Polynomial regression with Ridge regularization gave good cross-validation results for both datasets:

- **var1 (steam turbine):** degree 5, $\alpha = 2.8480$, CV MSE = 0.4730, CV $R^2 = 0.9484$, and training $R^2 = 0.9788$.
- **var2 (thermal reservoir):** degree 12, $\alpha = 0.1520$, CV MSE = 0.2590, CV $R^2 = 0.9949$, and training $R^2 = 0.9966$.

The results show why both polynomial degree and regularization strength matter. A model that is too simple misses important patterns, while an overly complex model can be less reliable. Cross-validation was the main basis for choosing the final degrees. Since the true test labels were not provided, the final test prediction error remains unknown.